COURSE NLP301 / MODULE NL

Text Processing Fundamentals

Learning goal: explain core concepts, compare related methods, and apply them to realistic machine learning or educational technology scenarios.

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01. What is NLP?

Natural language processing builds methods for working with human language, including classification, search, extraction, translation, and question answering. Meaning depends on context, domain, and the language used. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

02. What is tokenization?

Tokenization breaks text into words, subwords, or other units the model can process. A word tokenizer might split on punctuation, while modern language models often use subwords so rare words can be assembled from familiar pieces. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

03. What are sentence tokenization and text cleaning?

Sentence tokenization identifies sentence boundaries; cleaning may normalize whitespace and encoding. Preserve original source text and offsets for citations. Aggressive lowercasing or punctuation removal may destroy useful meaning. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

04. When should stop words be removed?

Stop words are frequent words such as 'the'. Removing them can help some sparse retrieval pipelines, but words such as 'not' can reverse meaning. For modern contextual models, retain natural text unless tests show a benefit. The choice is context dependent. The right approach depends on the objective, the data distribution, the cost of errors, and operational constraints such as latency, memory, interpretability, and the amount of labeled data available. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

05. How do stemming and lemmatization differ?

Stemming applies mechanical rules and can produce nonwords; lemmatization maps an inflected word to a dictionary form using linguistic information. Neither should automatically replace the original text needed for readable evidence. A useful way to understand how do stemming and lemmatization differ is to compare the methods not only by definition, but also by the kind of input they expect, the output they produce, and the situations in which each is useful. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

06. What are n-grams?

An n-gram is a sequence of n adjacent units. Word bigrams can preserve phrases such as 'gradient descent' that single-word counts miss. Larger n-grams grow sparse quickly and need more data. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

07. What is an out-of-vocabulary token?

An OOV token was absent from a model's fixed vocabulary. Older pipelines map it to UNK; subword tokenizers can often decompose it instead. Domain-specific terms can still be represented poorly and should be evaluated. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.